Sleep and the Daytime Course of Anhedonia in Adolescent Depression: An Ecological Momentary Assessment Study

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Abstract

Background: Mounting evidence suggests that sleep disturbances and depression are linked through disruptions in reward processing. This pathway may be especially relevant to adolescent depression, as adolescence is characterized by more frequent sleep disturbances, reduced sleep, and refinement of reward circuitry.

Aim: This preregistered study uses high-density ecological momentary assessment (EMA) data from the Characterization and Treatment of Adolescent Depression Studies (CTADS) cohort (collected from 2023 to 2026) to test sleep disturbances and duration as predictors of anticipatory and consummatory anhedonia and to determine whether effects exhibit diurnal variation.

Methods: Sixty-five adolescents (12-18 years; 43 female, 22 male; 41 with major depressive disorder, 24 healthy volunteers) completed 120 EMA prompts over 30 days. Participants reported prior-night sleep disturbances and duration once per day, and anticipatory and consummatory anhedonia four times per day. We ran two Bayesian multilevel linear models, estimating anticipatory and consummatory anhedonia, respectively, as functions of prior-night sleep disturbances, time of day, diagnostic group, and their interactions. We then ran two exploratory models assessing sleep duration in the same manner.

Results: In adolescents with MDD, whereas self-reported sleep disturbances did not predict anhedonia, longer sleep duration predicted both lower anticipatory and lower consummatory anhedonia. In healthy adolescents, greater sleep disturbances predicted lower anticipatory and lower consummatory anhedonia. Effects of sleep disturbances on anhedonia tended to increase across the day in adolescents with MDD and healthy adolescents.

Conclusions: Sleep disturbances and duration differentially predict next-day anhedonia in adolescents with MDD. For adolescent MDD patients, treatments that target sleep duration, as opposed to sleep disturbances, could provide more immediate effects on anhedonia. Time-of-

day moderation effects suggest the mechanisms through which sleep disturbances (but not sleep duration) affect anhedonia may involve diurnal biological processes.

General Scientific Summary

In adolescents with MDD, whereas self-reported sleep disturbances did not predict anhedonia, longer sleep duration predicted both lower anticipatory and lower consummatory anhedonia. Moderation results suggest that the mechanisms through which sleep disturbances affect anhedonia may involve diurnal biological processes.

Sleep and the Daytime Course of Anhedonia in Adolescent Depression: An Ecological Momentary Assessment Study

In the United States, approximately 19% of adolescents aged 12–19 meet criteria for major depressive disorder (MDD; Brody & Hughes, 2025). Sleep disturbances—an overarching term to describe a variety of problems, including insomnia, frequent nighttime waking, low sleep efficiency, and light sleep—impact at least two-thirds of adolescents with MDD (Zhao et al., 2024; see Lovato & Gradisar, 2014). Sleep disturbances are a common symptom of depression, but they also contribute to syndrome onset (Scott et al., 2021), interfere with treatment (Troxel et al., 2012), and persist after other depressive symptoms improve (Clarke & Harvey, 2012), which may increase risk of depression recurrence. These relations between sleep disturbances and depression may be partly explained by an indirect pathway involving aberrant reward processing (see the Integrated Sleep and Reward Model; Boland et al., 2020). This pathway may be especially consequential during adolescence, as adolescence is characterized by more frequent sleep disturbances and reduced sleep duration (Kansagra et al., 2020; Morrison et al., 1992), refinement of reward circuitry (Walker et al., 2017), and heightened depression risk (Thapar et al., 2012). In the current study, we investigate this putative pathway using high-density ecological momentary assessment (EMA) data to examine the effects of sleep disturbances and duration on anhedonia: a subjective indicator of aberrant reward processing and a core symptom of adolescent MDD.

Research examining between-person effects of sleep on the subjective experience of reward reliably finds that individuals with greater sleep disturbances report higher anhedonia contemporaneously. In a study of 166 Finnish adolescent psychiatric outpatients with unipolar depression, those with multiple sleep problems reported higher anhedonia and more suicidal thoughts than those with no or minor sleep problems; the groups did not differ in reports of any other depressive symptom (Urrila et al., 2012). Likewise, in a study by Wang and colleagues (2025a), 200 adolescent psychiatric outpatients with MDD completed questionnaires assessing

depression, anhedonia, and sleep quality at a single time point. The authors used network analysis to construct a sleep quality network and a co-occurrence network with anhedonia; in the co-occurrence network, sleep disturbances had a prominent bridging connection with anhedonia. In another study of 200 adolescents with MDD, sleep disorder was contemporaneously associated with higher anhedonia, even when controlling for the mediating role of life stress (Zhao et al., 2024).

To our knowledge, only two studies have examined within-person effects of sleep on anhedonia in young people. Both found that on nights that individuals reported more sleep disturbances or shorter sleep duration, they tended to report higher next-day anhedonia; although, effects differed for anticipatory versus consummatory anhedonia. In a study by Wieman and colleagues (2022), over one week, 260 unselected undergraduates completed daily diaries assessing sleep, reward responsiveness, depression, and anhedonia. At the day level, poorer sleep quality, shorter sleep duration, and longer awakening after sleep onset predicted blunted anticipatory and consummatory reward responsiveness, while increased sleep onset latency and lower sleep efficiency predicted blunted consummatory reward responsiveness only. In another study (Patel et al., 2025), 48 adolescents with mixed diagnoses and high suicide risk completed 28 daily diaries assessing perceived sleep onset latency and sleep duration as well as EMA prompts assessing anhedonia 3-6 times per day. The authors found that within individuals, longer sleep onset latency was associated with higher consummatory anhedonia (although this finding was inconsistent across models) and was not associated with anticipatory anhedonia. Consistently, within individuals, longer sleep duration predicted lower consummatory anhedonia but not anticipatory anhedonia. In addition to demonstrating that sleep impacts next-day anhedonia, these results suggest a functional distinction between anticipatory and consummatory anhedonia, supporting the separate examination of these temporal domains of reward processing.

There are two patent gaps in the literature. First, as noted, to our knowledge, only two studies (Wieman et al., 2022; Patel et al., 2025) have examined within-person effects of sleep on anhedonia, and only one (Patel et al., 2025) did so in a clinical sample of adolescents. Second, no studies have examined whether or how effects of sleep on anhedonia might vary by time of day. Identifying when sleep most strongly influences anhedonia could suggest whether the effects of sleep are driven by circadian rhythms, accumulated sleep pressure, or other time-varying cognitive or affective processes. Additionally, if associations exhibit diurnal variation, interventions targeting sleep or anhedonia may be more effective when delivered at particular times.

The Current Study

We leveraged data from a high-density EMA study to test sleep disturbances and duration as predictors of anticipatory and consummatory anhedonia in adolescents with and without MDD, and to determine whether effects exhibit diurnal variation. We hypothesized that within individuals, sleep disturbances would positively predict anhedonia and that this effect would be stronger in adolescents with MDD compared to healthy adolescents. We also hypothesized that within individuals, time would moderate the effect of sleep disturbances on anhedonia, such that the effect would weaken across the day. We expected the moderating effect of time would weaken less across the day in adolescents with MDD compared to healthy adolescents. In addition to our preregistered analyses pertaining to sleep disturbances, we conducted exploratory analyses pertaining to sleep duration, for which we did not pose a priori hypotheses.

Methods

This preregistered study utilized data from the Characterization and Treatment of Adolescent Depression Studies (CTADS) cohort (National Institute of Mental Health Institutional Review Board protocol 18-M-0037; ClinicalTrials.gov NCT03388606; collected from 2023 to 2026). Only methods relevant to the current study are described below. Data collection for

CTADS is ongoing; the current study includes data collected as of May 21, 2026: the date of preregistration.

Participants and Recruitment Procedures

Adolescents aged 11-17 with and without MDD were recruited from Washington, D.C. and surrounding areas through social media advertisements, paper advertisements posted in the community, mailings to local zip codes, and community referrals. Participants were ineligible for the study if they had an estimated IQ<70; regularly used substances within the previous three months; had ever experienced cardinal bipolar disorder symptoms; met diagnostic criteria for posttraumatic stress disorder, a psychotic disorder, or moderate-to-severe autism spectrum disorder; or had a neurological disorder or unstable medical condition. Before data collection, adolescents provided informed, written assent and parents/guardians provided informed, written consent. Following data collection, two participants were excluded from analyses because they completed <25% of EMA prompts (a preregistered exclusion criterion ensuring each participant contributed enough observations to reliably estimate within-person effects).

Our final sample consisted of 65 adolescents (43 female, 22 male; *M*=15.96 years, *SD*=1.39, range=12.75-18.67): 41 who met DSM-5 criteria for MDD within the six months prior to enrollment (as determined using the Kiddie Schedule for Affective Disorders and Schizophrenia [K-SADS] interview; Kaufman et al., 1997), and 24 healthy adolescents. The sample was 49.23% White, 13.85% Black/African American, 6.15% Asian, 1.53% American Indian/Alaskan Native, and 24.62% biracial/multiracial; 4.62% of participants did not report race. The sample was 13.85% Hispanic/Latino; 81.54% of participants were neither Hispanic nor Latino, and 4.62% did not report ethnicity. Most adolescents (44.62%) were from families with an annual household income over $150,000, followed by those with incomes of $125,000-149,999 (10.77%), 100,000-124,999 (7.69%), 75,000-99,999 (6.15%), 50,000-74,999 (4.62%), and 25,000-49,000 (3.08%); 26.15% did not report income. On average, adolescents completed 75.45% of EMA prompts (*M*=90.54, *SD*=22.67) over 28.51 days.

Procedures

Following informed consent, adolescents were trained on the high-density EMA protocol. Training involved meeting with a research assistant to review EMA items and practice responding to prompts using the m-Path software (Mestdagh et al., 2023). Participants who did not own a smartphone received a study-provided smartphone configured solely for EMA data collection.

Adolescents were asked to complete EMA prompts four times a day (before school, during school, in the afternoon, and before bed) for 30 days. At each prompt, adolescents were asked to rate their experiences since the previous prompt. Prompts were scheduled pseudorandomly, within 1-hour timeslots tailored to each adolescent's schedule to increase feasibility and compliance (e.g., during lunchtime at school). When requested, prompt timeslots were adjusted to accommodate changes in adolescents' schedules. All participants received a check-in during the third week of the EMA protocol to encourage continued participation.

Adolescents were compensated up to $400 for the 30-day study period. Compensation was distributed week-by-week based on EMA compliance rate for that week (standard compensation plus bonus payment for compliance >80%; standard compensation for compliance=50-80%; no compensation for compliance <50%). The payment increased each week to encourage maintenance.

Measures

Sleep variables were assessed using two EMA items: *I had problems sleeping* (sleep disturbances) and *I slept…* (sleep duration). Participants rated sleep disturbances on a sliding visual analogue scale (VAS) with underlying (unseen) values ranging from 0-100, where 0=*not at all* and 100=*extremely*. Participants rated sleep duration on a similar VAS, where 0=*too little*, 50=*just right*, and 100=*too much*. Participants reported on their sleep only during their first EMA response of the day (e.g., if a participant did not respond to the first EMA prompt, the sleep items were presented during the second prompt of the day).

Anhedonia was assessed using two EMA items: *I looked forward to things* (anticipatory) and *I enjoyed what I was doing* (consummatory). Adolescents rated each item on a sliding VAS, where 0=*not at all* and 100=*extremely*. Participants reported anhedonia since the previous prompt at every prompt (four times per day). Prior to analyses, anticipatory and consummatory anhedonia were reverse-scored so that higher values reflected higher anhedonia.

Statistical Analyses

We conducted all analyses in R version 4.5.1 (R Core Team, 2025), primarily using the *brms* package (Bürkner, 2017). For each model described below, four Markov chains were run, each with 4,000 iterations. Package-default, weakly informative priors were used for all fixed effects and variance components. Sleep disturbances, sleep duration, anticipatory anhedonia, and consummatory anhedonia were person-mean centered prior to analyses.

Preregistered Analyses

We ran two Bayesian multilevel linear models (one modeling anticipatory anhedonia and one modeling consummatory anhedonia), with EMA reports (level 1) nested within days (level 2) nested within individuals (level 3). In each model, anhedonia was estimated as a function of prior-night sleep disturbances (level 2), time of day (level 1), diagnostic group (level 3), and all interactions. We then re-ran the models including sex (level 3) and age (level 3) as covariates. All models included random intercepts for participants and for EMA day nested within participants, and random slopes for time, sleep disturbances, and their interaction across participants, with correlations among all participant-level random effects estimated by default.

Exploratory Analyses

We ran exploratory analyses investigating the effects of sleep duration on anticipatory and consummatory anhedonia, using the same model specifications as in our preregistered analyses, only substituting sleep disturbances with sleep duration.

Inference

Inference was based primarily on posterior distributions and 95% credible intervals for model parameters. We considered posterior probabilities ranging from 0.90 to 0.95 as moderate evidence of effects, and posterior probabilities exceeding 0.95 as strong evidence of effects.

Transparency and Openness

Our sample, hypotheses, and analytic plan were preregistered on Open Science Framework at https://osf.io/r685d/. Data (for *n*=61 participants who consented to public data sharing) and code used for statistical analyses are publicly available on GitHub at https://github.com/HannahEveMarshall/ctads_sleep_anhedonia. We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study.

Results

For all regression coefficients reported below, $\hat{R}$ values were between 1.00 and 1.01; bulk and tail effective sample sizes exceeded 1968 and 650, respectively. For descriptive statistics and intraclass correlation coefficients, see Table S1. For posterior correlations between variables, see Table S2.

Preregistered Analyses

Full model estimates are presented in Tables S3-4. Controlling for sex and age did not alter central findings; for results from analyses including these covariates, see Tables S11-12.

Anticipatory Anhedonia

We hypothesized that in adolescents with and without MDD, greater sleep disturbances would predict higher next-day anticipatory anhedonia. Contrary to predictions, in adolescents with MDD, we did not find evidence of perceived sleep disturbances predicting anticipatory anhedonia, β=0.101, 95% CrI[-0.091, 0.296], SE=0.099, P(β>0)=0.842. In healthy adolescents, however, we found moderate evidence that sleep disturbances predicted lower anticipatory anhedonia (opposite to the hypothesized effect direction), β=-0.272, 95% CrI[-0.684, 0.136], SE=0.209, P(β<0)=0.902. We expected sleep disturbances to more positively predict anhedonia

in adolescents with MDD compared to healthy adolescents; we found strong evidence of this hypothesized between-group difference, β=-0.371, 95% CrI[-0.802, 0.068], SE=0.226, P(β<0)=0.952.

We also predicted that time would moderate the effect of sleep disturbances on anticipatory anhedonia, such that the effect would weaken across the day. Time of day did not moderate the positive-trending effect of sleep disturbances in adolescents with MDD, β=-0.004, 95% CrI[-0.024, 0.015], SE=0.010, P(β<0)=0.662; however, in healthy adolescents, the negative effect of sleep disturbances became more positive across the day, β=0.030, 95% CrI[-0.012, 0.072], SE=0.021, P(β>0)=0.923. There was moderate evidence of a group difference, β=0.034, 95% CrI[-0.010, 0.078], SE=0.023, P(β>0)=0.933.

Consummatory Anhedonia

Our hypotheses regarding consummatory anhedonia were identical to those for anticipatory anhedonia. Contrary to predictions, in adolescents with MDD, we did not find credible evidence of sleep disturbances predicting consummatory anhedonia, β=-0.104, 95% CrI[-0.289, 0.101], SE=0.102, P(β<0)=0.840. In healthy adolescents, however, we found moderate evidence that sleep disturbances predicted lower consummatory anhedonia (opposite to the hypothesized effect direction), β=-0.292, 95% CrI[-0.690, 0.112], SE=0.206, P(β<0)=0.922. We did not find credible evidence of a group difference, β=-0.207, 95% CrI[-0.660, 0.252], SE=0.230, P(β<0)=0.816.

We expected time to moderate the effect of sleep disturbances on anhedonia, such that the effect would weaken across the day. We found strong evidence that in adolescents with MDD, across the day, the negative-trending effect of sleep disturbances became more positive, β=0.018, 95% CrI[-0.003, 0.036], SE=0.010, P(β>0)=0.958. Similarly, in healthy adolescents, we found moderate evidence that the negative effect of sleep disturbances became more positive across the day, β=0.033, 95% CrI[-0.008, 0.074], SE=0.021, P(β>0)=0.941. We did not

find credible evidence of a three-way interaction between sleep disturbances, time, and diagnostic group, β=0.017, 95% CrI[-0.030, 0.064], SE=0.023, P(β>0)=0.776.

[FIGURE 1]

Exploratory Analyses

Full model estimates are presented in Tables S5-6. Controlling for sex and age did not alter central findings; for results from exploratory analyses including these covariates, see Tables S13-14.

Anticipatory Anhedonia

In our parallel analyses of sleep duration, we found moderate evidence that in adolescents with MDD, more sleep predicted lower next-day anticipatory anhedonia, β=-0.194, 95% CrI[-0.480, 0.090], SE=0.145, P(β<0)=0.910. There was further moderate evidence of a group difference, β=0.414, 95% CrI[-0.165, 0.983], SE=0.292, P(β>0)=0.922; in healthy adolescents, sleep duration did not predict anticipatory anhedonia, β=0.225, 95% CrI[-0.262, 0.720], SE=0.249, P(β>0)=0.818. Effects did not vary by time of day.

Consummatory Anhedonia

Our findings for consummatory anhedonia were consistent with those for anticipatory anhedonia. We found strong evidence that in adolescents with MDD, more sleep predicted lower consummatory anhedonia, β=-0.312, 95% CrI[-0.580, -0.036], SE=0.139, P(β<0)=0.987. There was moderate evidence that this effect varied by group, β=0.380, 95% CrI[-0.164, 0.945], SE=0.281, P(β>0)=0.914; in healthy adolescents, sleep duration did not predict consummatory anhedonia, β=0.072, 95% CrI[-0.394, 0.539], SE=0.242, P(β>0)=0.611. Effects did not vary by time of day.

[FIGURE 2]

Discussion

The current study utilized 30 days of EMA data from adolescents with and without MDD to test sleep disturbances and duration as predictors of anticipatory and consummatory anhedonia, and to determine whether effects exhibited diurnal variation. Self-reported sleep disturbances did not meaningfully impact next-day anhedonia: Sleep disturbances did not predict anticipatory or consummatory anhedonia in adolescents with MDD, and sleep disturbances predicted *lower* anticipatory and consummatory anhedonia in healthy adolescents. With respect to sleep duration, however, more sleep predicted lower anticipatory and lower consummatory anhedonia specifically in adolescents with MDD. Time-of-day moderation effects suggest that the mechanisms through which sleep disturbances (but not sleep duration) affect anhedonia may involve diurnal biological processes.

Self-Reported Sleep Disturbances May Not Meaningfully Impact Next-Day Anhedonia

Contrary to our hypotheses, sleep disturbances did not predict anticipatory or consummatory anhedonia in adolescents with MDD. This may reflect a cumulative effect of sleep disturbances on anhedonia that is not observable at the day level. For example, for an adolescent who usually experiences sleep disturbances, a single night of high-quality sleep may not be sufficient to improve next-day symptoms; instead, symptoms may only be impacted after multiple consecutive nights of sleep. The notion that symptoms may only improve after an enduring reduction in sleep disturbances is consistent with evidence that adolescents with MDD exhibit greater emotional inertia (their emotional states tend to persist and are difficult to change; e.g., Liu et al., 2026).

More Sleep Predicts Lower Anhedonia Only in Adolescents with Depression

Building on existing literature in healthy adults (Wieman et al., 2022), this is the first study to demonstrate in adolescents with MDD that getting more sleep predicts next-day

decreases in both anticipatory and consummatory anhedonia. In the current sample, most adolescents with MDD reported getting too little sleep on average (see Table S1). Reduced sleep is associated with diminished effort expenditure in pursuit of rewards, which can result in individuals experiencing less pleasure in the moment and anticipating less future pleasure (Boland et al., 2020; Treadway et al., 2012). Sleep deficits are also known to impair emotion regulation (e.g., Lollies et al., 2022), meaning when adolescents get less sleep, they may have more difficulty upregulating positive affect the next day. These alterations may occur because sleep duration constrains the time available for nocturnal processes implicated in depression. For example, reduced sleep limits the opportunity for synaptic homeostasis within reward and affective regions, which may contribute to the pathophysiology of depression (Wang et al., 2025b). Similarly, not getting enough sleep can disrupt the consolidation of emotional memory (Niu et al., 2025; Bennion et al., 2015; Hutchison & Rathore, 2015), impairing individuals' ability to accurately valuate (and, in turn, seek out) potential future rewards based on past pleasurable experiences.

Contrarily, in healthy adolescents, self-reported sleep duration was not associated with next-day anhedonia. This may be because healthy adolescents lack the underlying vulnerabilities associated with MDD. MDD is characterized by altered reward processing and dysregulation in frontolimbic circuitry involved in emotion regulation (e.g., Miller et al., 2015; Zhang et al., 2026). For adolescents with MDD, reduced sleep may further compromise these systems (by limiting the time available for nocturnal process, as described above), leading to increases in anhedonia; however, for healthy adolescents without these pre-existing aberrations, the effects of reduced sleep on these systems may not be pronounced enough to impact anhedonia the following day.

The Effects of Sleep Disturbances (But Not Sleep Duration) May Involve Diurnal Biological Processes

Findings concerning the moderating effects of time of day suggest that the mechanisms through which sleep disturbances (but not sleep duration) affect anhedonia may involve diurnal biological processes. The associations between sleep disturbances and consummatory anhedonia increased across the day in adolescents with MDD and healthy adolescents, and the same was found for anticipatory anhedonia in healthy adolescents. The fact that sleep disturbances were more positively associated with anhedonia later in the day suggests that accumulated sleep pressure, fatigue, or related circadian processes may be implicated. Conversely, time did not moderate the effects of sleep duration in either group, supporting the notion that the effects of sleep duration may be related to nocturnal processes, such as synaptic homeostasis or consolidation of emotional memory, as opposed to diurnal biological processes.

Limitations and Future Directions

Several considerations should be made when interpreting the present findings and identifying directions for future research. First, the current study only employs subjective measures of sleep; future work should integrate subjective and objective measures to facilitate a broader investigation of sleep effects. Second, while the current study examines whether sleep variables predict anhedonia, these effects are likely bidirectional. Putative bidirectional processes should be examined in future work using methods like dynamic structural equation modeling, which are well-suited to bidirectional and autoregressive frameworks. Additionally, in the current sample, healthy adolescents reported a relatively low incidence of sleep disturbances (see Table S1), which may have introduced floor effects. To better understand whether findings are unique to adolescents with MDD, future researchers may wish to include a sample of adolescents with significant sleep disturbances who are otherwise healthy.

Constraints on Generality

More female than male adolescents enrolled in the study, consistent with the higher prevalence rate of MDD in adolescent females relative to males (Brody & Hughes, 2025). The racial composition of the current sample was broadly representative of the U.S. population. While this enhances the generalizability of findings to the broad population, findings may not generalize to racial groups that are underrepresented nationally, such as American Indians and Alaskan Natives.

Conclusion

To our knowledge, the present study is one of only a few to examine person-specific, within-day effects of sleep on anhedonia, and is the first to examine these effects specifically in adolescents with MDD. We found that more sleep consistently predicted lower anticipatory and consummatory anhedonia in adolescents with MDD. In contrast, sleep disturbances were not meaningfully associated with next-day anhedonia. Thus, for adolescent MDD patients, treatments that target sleep duration, as opposed to sleep disturbances, could provide more immediate effects on anhedonia. Notably, associations between sleep disturbances (but not sleep duration) and anhedonia varied across the day, suggesting that the mechanisms through which sleep disturbances affect anhedonia may involve diurnal biological processes. More broadly, these findings provide a potentially fruitful direction for future research on adolescent depression and anhedonia, particularly research incorporating complementary, objective sleep measures.

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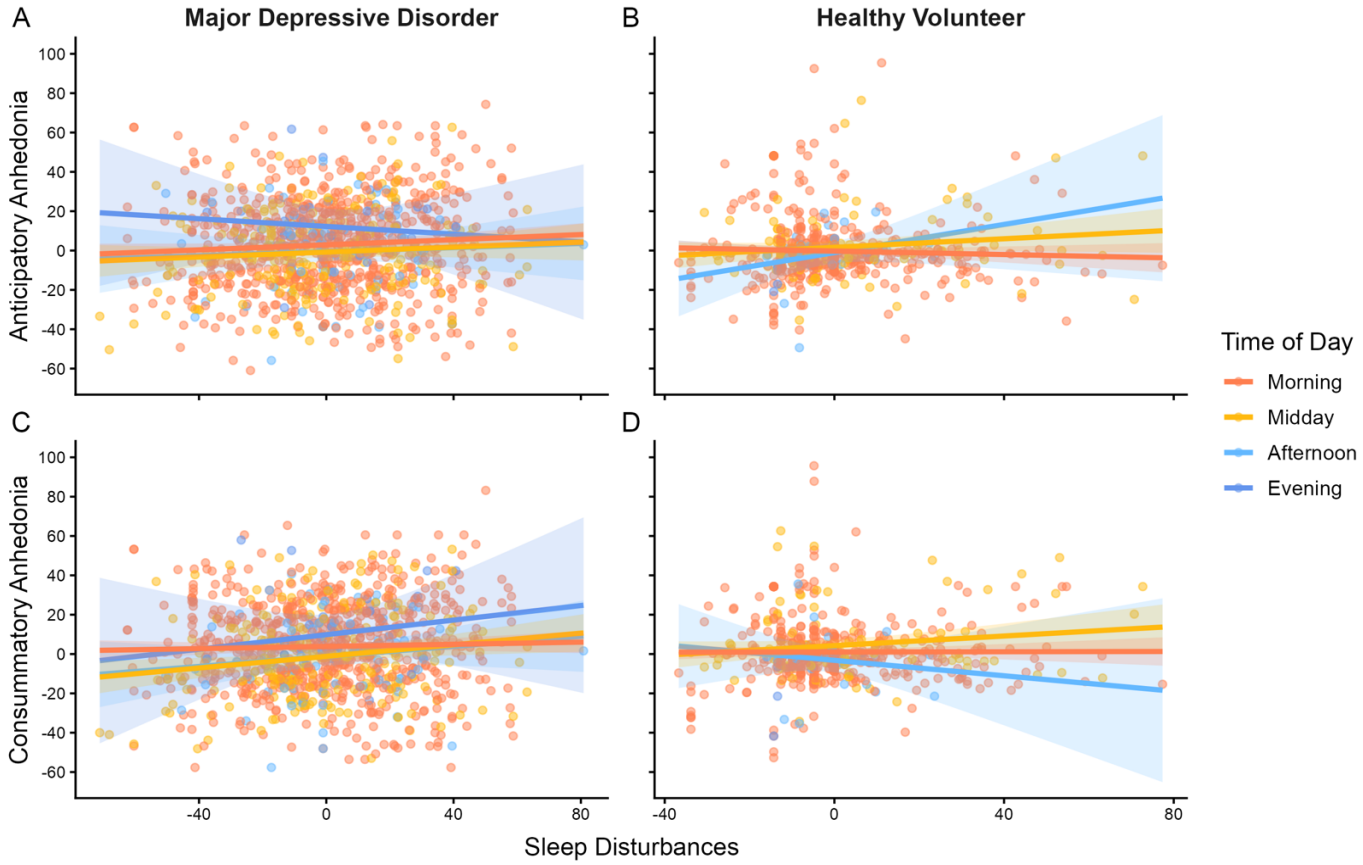
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Figures

Figure 1

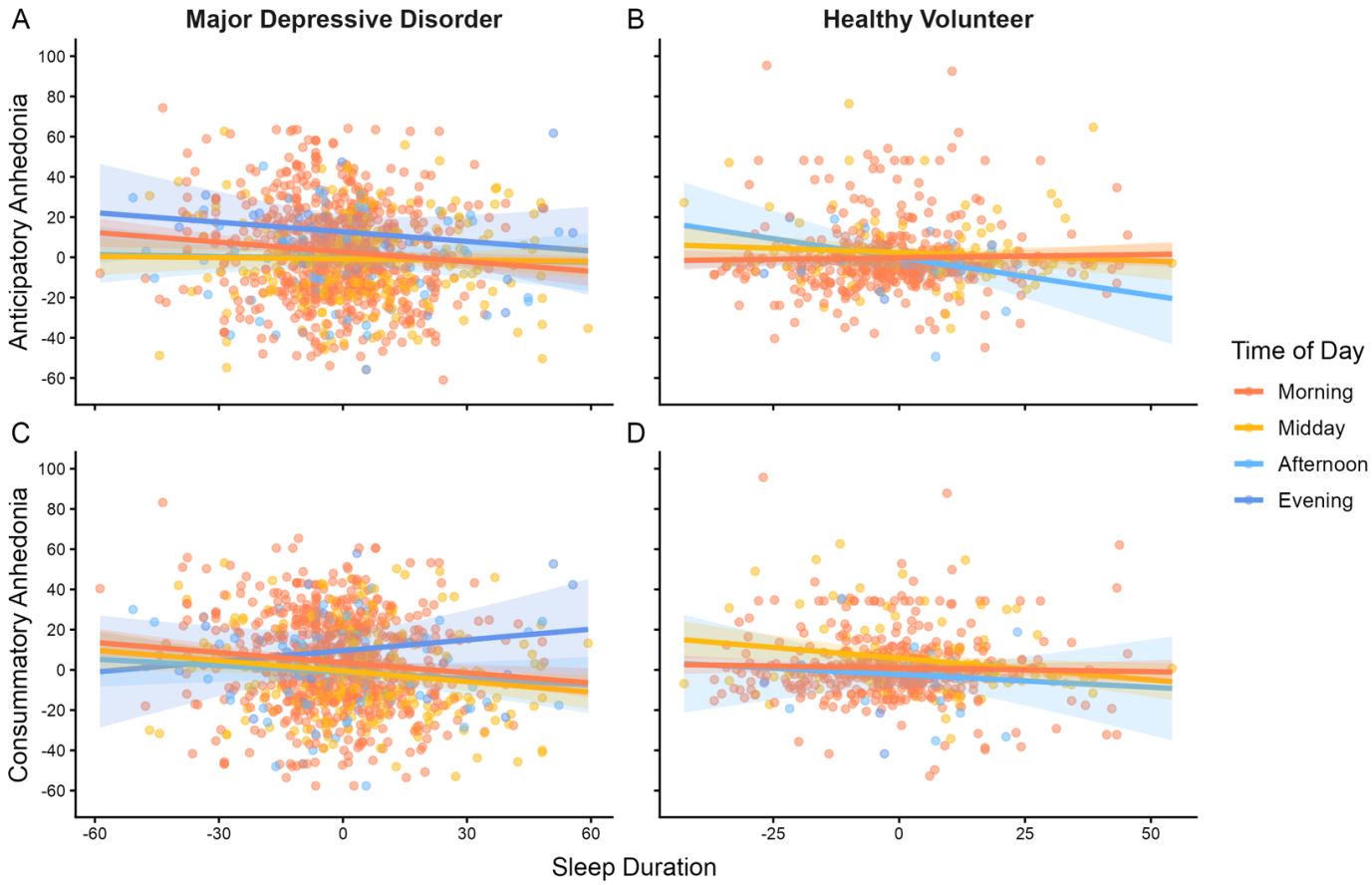
Next-Day Anhedonia by Sleep Disturbance and Time of Day



Note. Anhedonia and sleep disturbances were person-mean centered. Time-of-day bins were based on naturally occurring local minimums in the kernel density, averaged across diagnostic groups (before 10:15 AM, 10:15 AM–2:31 PM, 2:31–7:07 PM, after 7:07 PM). Ribbons indicate standard errors. Evening trend lines have been omitted from healthy volunteer plots; due to the dearth of evening reports in the HV group, evening effects could not be reliably estimated.

Figure 2

Next-Day Anhedonia by Sleep Duration and Time of Day



Note. Anhedonia and sleep duration were person-mean centered. Time-of-day bins were based on naturally occurring local minimums in the kernel density, averaged across diagnostic groups (before 10:15 AM, 10:15 AM–2:31 PM, 2:31–7:07 PM, after 7:07 PM). Ribbons indicate standard errors. Evening trend lines have been omitted from healthy volunteer plots; due to the dearth of evening reports in the HV group, evening effects could not be reliably estimated.